COURSE-0000 — Institution

Course Title

Lecture Notes

Your Name

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Lesson 1

Using this template

Lecture	Lesson 1 (Introduction)
Textbook	H&P [1] §1.1–1.3
Keywords	sidenotes, index, bilingual build

This chapter is a showcase of every macro the class provides. Delete it when you start a real course, but keep `02-lesson-skeleton.tex` as a starting point for new lessons.

1.1 Margin notes

The right-hand column is for anything that supports the main text without interrupting it. Numbered remarks use `\sidenote`[1] and unnumbered remarks use `\margin`.

Three labelled variants make the *kind* of side content visible at a glance:

- `\source{...}` — where the material comes from (lecture, textbook section, paper);
- `\tip{...}` — practical advice;
- `\caution{...}` — pitfalls and misconceptions.

[1] Like this one. Numbers restart at every chapter.

An unnumbered margin note.

Source Amdahl's original paper [2]; H&P §1.9.

Tip Use `\tip` for exam-relevant advice or mnemonics.

Caution Use `\caution` for common misconceptions.

1.2 Terms and the index

A *pipeline* is introduced with `\term`, which emphasizes the word, adds it to the index, and prints a gloss in the margin (the Japanese term) unless the edition sets the class option `gloss=false`, as this English edition does. Later mentions of a *pipeline* use `\term*` so the index does not fill up with repeated page numbers. Terms can be nested: *speedup* and *Amdahl's law* both go to the index.

Definition 1.1 (Speedup). For a change that makes a fraction f of the execution time s times faster, the overall speedup is

$$S = \frac{1}{(1 - f) + f/s}. \tag{1.1}$$

Example 1.2. If $f = 0.9$ and $s \to \infty$, Equation (1.1) gives $S = 10$: the untouched 10 % bounds the achievable speedup.

Note 1.3. `definition`, `example` and `note` share one counter per chapter, so they can be cross-referenced with `\cref`.

> **Key point — Amdahl's law**
> Optimize the common case; the part you do not touch limits the gain. Margin macros work inside this box too.

1.3 Figures, tables and code

Small figures go in the margin (Figure 1.1); normal figures use the standard `figure` environment; wide figures use `widefigure`, which spans the body and the margin column (Figure 1.3). Figures shared between the two editions live in `figures/` and switch their labels with `\iflangja`.

Code uses `listings`:

From inside a key point.

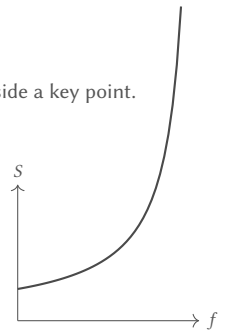


Figure 1.1: A figure in the margin (`marginfigure`).

Listing 1.1: A loop in assembly.

```
loop:   add   r1, r1, r2      ; accumulate
        addi  r3, r3, -1
```

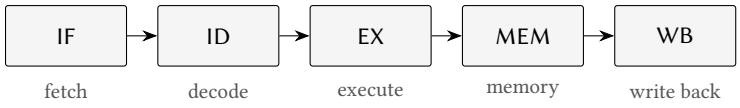


Figure 1.2: A classic five-stage pipeline (shared TikZ source).

Cycle	1	2	3	4	5	6	7
add	IF	ID	EX	MEM	WB		
sub		IF	ID	EX	MEM	WB	
lw			IF	ID	EX	MEM	WB

Figure 1.3: A wide float spanning body and margin (`widefigure`).

```
        bne   r3, r0, loop    ; branch if not done
```

Pseudocode uses `algorithmicx`:

Algorithm 1 Two-bit saturating counter update

if taken **then**
    $c \leftarrow \min(c + 1, 3)$
else
    $c \leftarrow \max(c - 1, 0)$
end if

1.4 Bilingual writing

Both editions share `format/`, `figures/` and `references.bib`; only the prose under `en/` and `ja/` differs. Inside shared files use `\iflangja{japanese}{english}` or the shorthands `\ja{...}` and `\en{...}`.

> **Key point — Lesson summary**
> - Margin macros carry sources, tips, cautions and asides.
> - `\term` defines a glossary term once; `\term*` marks later mentions.
> - Both editions share figures, bibliography and class.

Lesson skeleton

Lecture	Lesson 2, videos 1–12
Textbook	H&P [1] ch. 1
Keywords	first term, second term

Three to five sentences: what this chapter covers, why it matters, and how it continues the previous chapter.

2.1 First topic

The first sentence of the section. A *first term* is defined where the source defines it. The explanation follows the order of the source: what the concept is, why it is needed, and how it is used.

Source Lesson 2, videos 1–5; H&P §1.2.

Definition 2.1 (First term). A precise statement of the definition.

Example 2.2. A worked example with our own numbers.

Notation and units used in the example.

2.2 Second topic

Body text that builds on Section 2.1. Each margin item sits next to the sentence it supports, and a short run of text carries at most about three of them.

Source Lesson 2, videos 6–12.

A rule of thumb goes into a tip. The argument continues in the body, so that a reader who skips the margin still follows it. A longer aside that needs more than three lines goes into a numbered note.[1]

Tip A rule of thumb.

[1] Why the design exists, or a named real-world implementation, with its source.

> **Key point — Lesson summary**
> - The one thing to remember from the first section.
> - The one thing to remember from the second section.

Bibliography

[1] John L. Hennessy and David A. Patterson. *Computer Architecture: A Quantitative Approach*. 6th ed. Morgan Kaufmann, 2017.

[2] Gene M. Amdahl. "Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities". In: *AFIPS Conference Proceedings* 30 (1967), pp. 483–485.

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